Prozorro Competitiveness Predictor

Applied Data Science -- Project Write-Up

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Prediction Question

Given only the structural design of a Ukrainian government procurement tender -- its method type, publishing agency, sector, timing, and documentation profile -- can we predict whether the tender will attract only a single bidder? A single-bidder outcome is the dominant quantitative proxy for corruption risk in public procurement research (Fazekas & Kocsis, 2020). Ukraine's Prozorro system -- the world's most transparent national e-procurement platform -- publishes 28.7 million machine-readable tender records, and the full-scale Russian invasion of 2022 caused single-bidder rates to surge from ~50% to ~66%, making this prediction both urgent and policy-relevant.

Data Source

The raw data originates from the Prozorro Open Data API (public.api.openprocurement.org), which exposes the complete lifecycle of every Ukrainian government purchase in OCDS (Open Contracting Data Standard) JSON format. We synced the full dataset to a Google Cloud Storage bucket via an autonomous, self-terminating GCE Compute Engine instance, then loaded it into Snowflake using COPY INTO. The medallion architecture proceeds: Bronze (1.88M raw JSON records) -> Gold (feature-enriched with 101 columns including CPV sector flags, threshold proximity signals, and temporal markers) -> Platinum (4.5M competitive tenders with 34 engineered pre-tender features and a 10% daily-stratified training sample preserving temporal representativeness).

Architecture

Prozorro API -> GCE (paginated sync) -> GCS Bucket -> Snowflake Bronze -> Gold -> Platinum
Snowflake ML Registry <- LightGBM training (3 regime variants: peacetime, wartime, pooled)
Google Cloud Run (FastAPI backend) <- Model serving endpoint
Vite + React frontend (live dashboard) -> User browser

The pipeline spans three cloud services (GCP Compute Engine, GCP Cloud Storage, Snowflake) and three medallion layers. The trained LightGBM model is registered in Snowflake's ML Registry with version tracking, then loaded by a FastAPI backend deployed on Google Cloud Run that serves real-time predictions. The frontend is a Vite/React application that pulls live tenders from the Prozorro API, scores them against the model, and renders interactive risk visualizations.

Model Approach

We train LightGBM gradient boosted trees (Ke et al., 2017) across three temporal variants: peacetime (2016-2022), wartime (2022-2026), and pooled. The target is binary: single-bidder (1) vs. multi-bidder (0). All 34 features are strictly pre-tender to prevent target leakage. Hyperparameters: learning_rate = 0.03, num_leaves = 63, max_depth = 7, early stopping at 50 rounds. The wartime model achieves AUC = 0.772, F1 = 0.757, accuracy = 70.8% on held-out 2025-2026 data; the pooled model achieves AUC = 0.870 on peacetime holdout. A logistic regression baseline achieves only AUC = 0.707, confirming that the nonlinear interactions captured by gradient boosting are essential. SHAP analysis reveals that the invasion redistributed predictive signal from procedural features (tender period, documentation) onto structural features (procurement method, procurer identity) -- the central empirical finding. Cross-regime transfer shows a striking asymmetry: wartime -> peacetime transfers well (AUC = 0.77), but peacetime -> wartime collapses (AUC = 0.55).

Learnings

The hardest lesson was that data engineering consumed 70% of the project effort. Syncing 28.7M records from a rate-limited API required building autonomous cloud workers with checkpointing and self-termination -- infrastructure that felt distant from "data science" but proved essential. The most surprising finding was the cross-regime transfer asymmetry: a model trained on 2.3 years of wartime data outperforms one trained on 6 years of peacetime data, even on peacetime test sets. This taught me that distributional stress can improve generalization by forcing models to rely on robust structural features rather than fragile procedural ones. If I were to start over, I would invest earlier in the State Audit Service monitoring data (which provides confirmed violation labels) rather than relying solely on the single-bidder proxy.